



BobSim

TransientEval Lateral Transient Response

Step response and sustained-sine FRF

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Run Notes

- Representative step steer response
- Continuous sinusoidal responses
- Rise time uses 50 percent input to 90 percent response thresholds
- Frequency delays are reported at 0.5 Hz and 1.0 Hz
- Gain and phase slopes summarize the FRF trend
- Steering shown in degrees for readability

TransientEval Metrics Summary 15.0 m/s

Time Domain - Step Response

STEP RESPONSE

a_y Rise Time (50-90%)	0.08	s
Sideslip Rise Time (50-90%)	0.19	s
Yaw Rise Time (50-90%)	0.06	s
Yaw Peak Response Time	0.10	s
a_y DC Gain	33.66	$\frac{\text{m/s}^2}{\text{rad}}$
Sideslip DC Gain	0.031	rad/rad
Yaw DC Gain	2.22	(rad/s)/rad
Roll DC Gain	0.072	rad/rad

STEADY-STATE / OVERSHOOT

a_y Peak	3.46	m/s^2
a_y Steady-State	2.86	m/s^2
a_y Overshoot	21.0	%
Yaw Steady-State	0.191	rad/s
Yaw Overshoot	0.034	rad/s
Roll Steady-State	0.006	rad
Roll Overshoot	0.001	rad
Settling Time	1.30	s

TransientEval Metrics Summary 15.0 m/s

Frequency Domain - Sustained Sine

FREQUENCY RESPONSE - CORE

DC Gain (a_y)	33.66	$\frac{\text{m/s}^2}{\text{rad}}$
DC Gain (r)	2.22	(rad/s)/rad
Peak Gain (a_y)	34.85	$\frac{\text{m/s}^2}{\text{rad}}$
Peak Freq (a_y)	1.00	Hz
Bandwidth (-3 dB)	1.00	Hz
Gain Slope (a_y)	0.99	dB/dec
Gain Slope (r)	0.28	dB/dec
Phase Slope (a_y)	11.60	deg/dec
Phase Slope (r)	16.79	deg/dec

DELAY / COUPLING

Ay/Yaw Delay @ 0.5 Hz	0.015	s
Yaw/Steer Delay @ 0.5 Hz	0.028	s
Difference @ 0.5 Hz	-0.013	s
Ay/Yaw Delay @ 1.0 Hz	0.017	s
Yaw/Steer Delay @ 1.0 Hz	0.028	s
Difference @ 1.0 Hz	-0.011	s
Gain Variation	3.5	%
Fit Error (a_y)	3.46e-03	-
Fit Error (r)	9.89e-04	-

TransientEval Metrics Summary 20.0 m/s

Time Domain - Step Response

STEP RESPONSE

a_y Rise Time (50-90%)	0.10	s
Sideslip Rise Time (50-90%)	0.09	s
Yaw Rise Time (50-90%)	0.06	s
Yaw Peak Response Time	0.10	s
a_y DC Gain	53.66	$\frac{\text{m/s}^2}{\text{rad}}$
Sideslip DC Gain	-0.020	rad/rad
Yaw DC Gain	2.68	(rad/s)/rad
Roll DC Gain	0.118	rad/rad

STEADY-STATE / OVERSHOOT

a_y Peak	4.99	m/s^2
a_y Steady-State	4.57	m/s^2
a_y Overshoot	9.1	%
Yaw Steady-State	0.230	rad/s
Yaw Overshoot	0.029	rad/s
Roll Steady-State	0.010	rad
Roll Overshoot	0.000	rad
Settling Time	1.23	s

TransientEval Metrics Summary 20.0 m/s

Frequency Domain - Sustained Sine

FREQUENCY RESPONSE - CORE

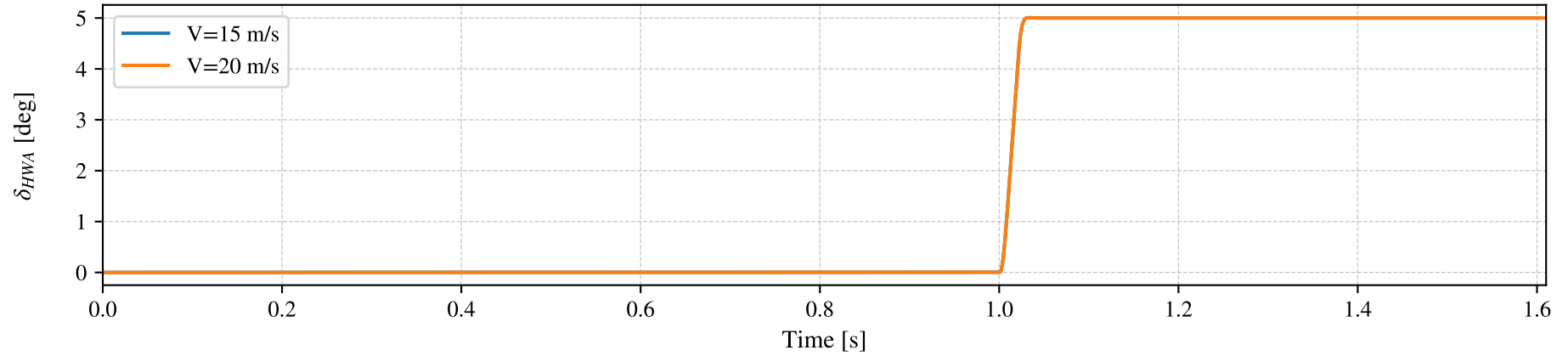
DC Gain (a_y)	53.66	$\frac{\text{m/s}^2}{\text{rad}}$
DC Gain (r)	2.68	(rad/s)/rad
Peak Gain (a_y)	53.66	$\frac{\text{m/s}^2}{\text{rad}}$
Peak Freq (a_y)	0.50	Hz
Bandwidth (-3 dB)	1.00	Hz
Gain Slope (a_y)	-0.02	dB/dec
Gain Slope (r)	0.06	dB/dec
Phase Slope (a_y)	25.96	deg/dec
Phase Slope (r)	19.05	deg/dec

DELAY / COUPLING

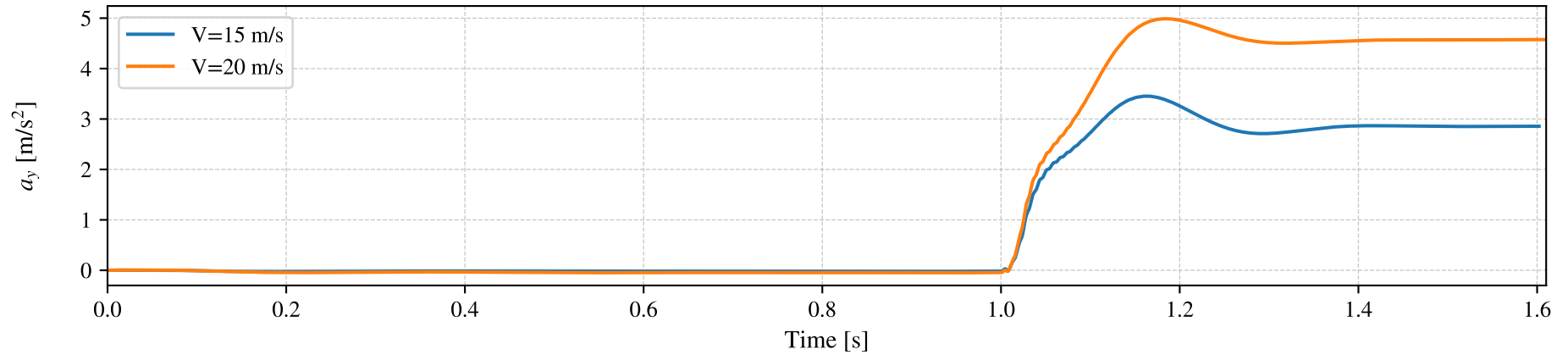
Ay/Yaw Delay @ 0.5 Hz	0.041	s
Yaw/Steer Delay @ 0.5 Hz	0.032	s
Difference @ 0.5 Hz	0.009	s
Ay/Yaw Delay @ 1.0 Hz	0.042	s
Yaw/Steer Delay @ 1.0 Hz	0.032	s
Difference @ 1.0 Hz	0.010	s
Gain Variation	0.1	%
Fit Error (a_y)	7.73e-03	-
Fit Error (r)	4.15e-03	-

Step Steer Response (TransientEval)

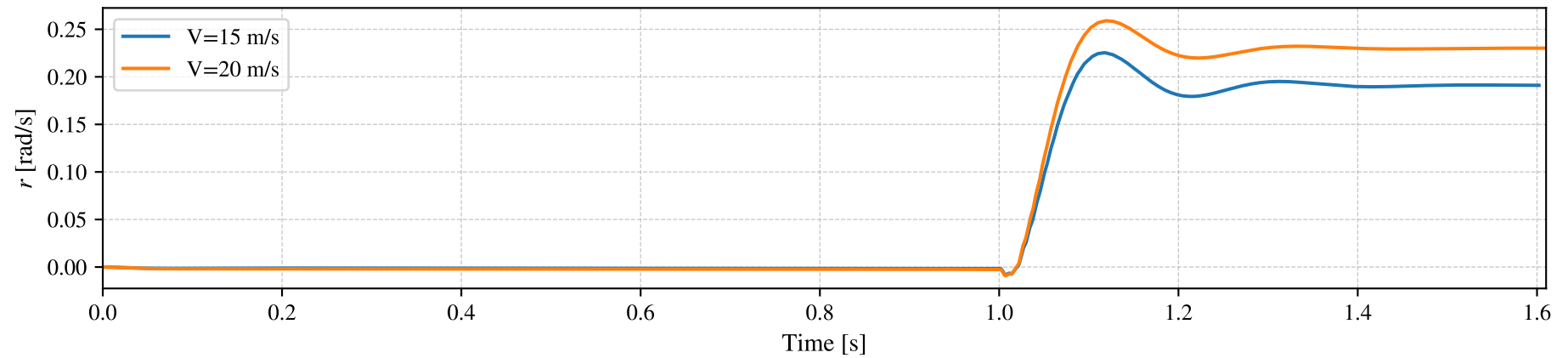
Steering Wheel Angle δ_H



Lateral Acceleration a_y

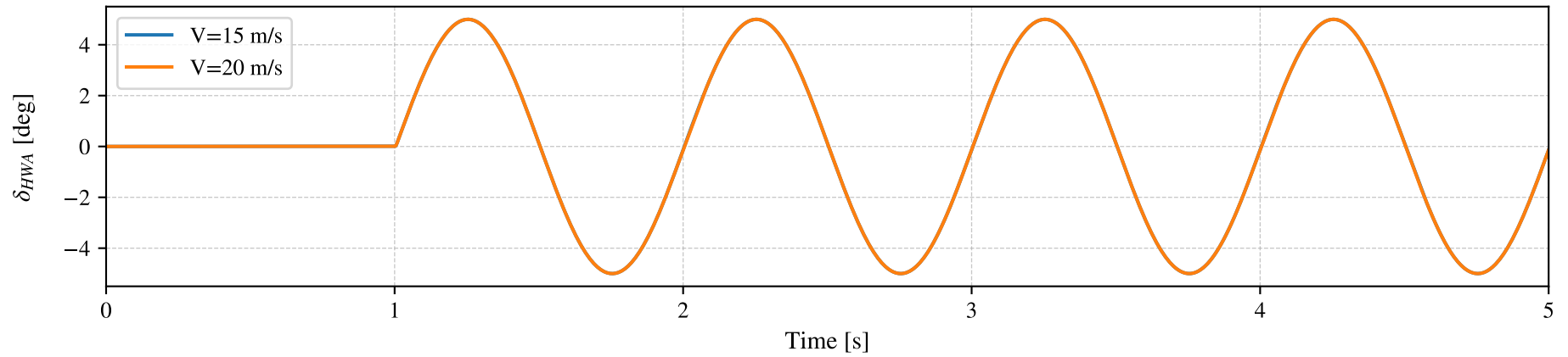


Yaw Velocity r

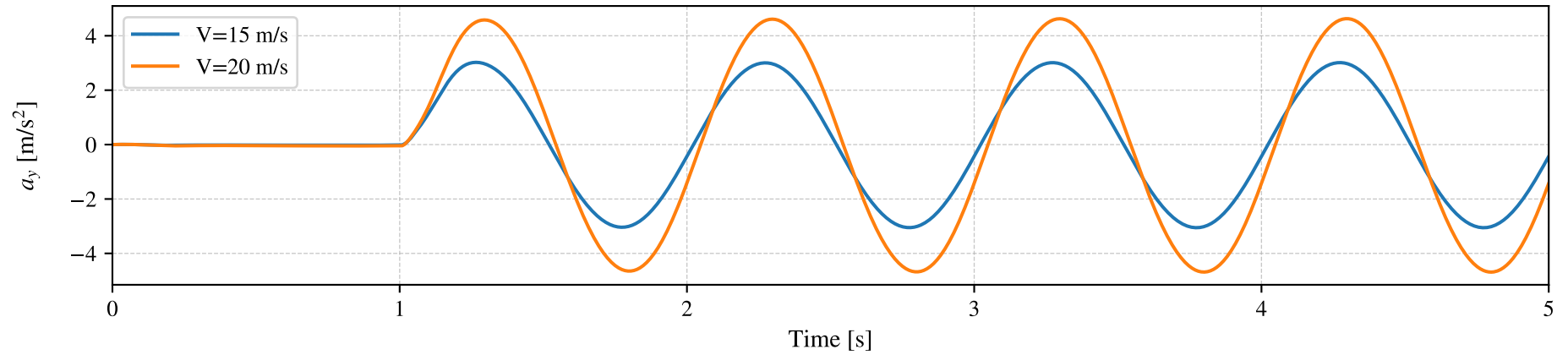


Continuous Sinusoidal Response

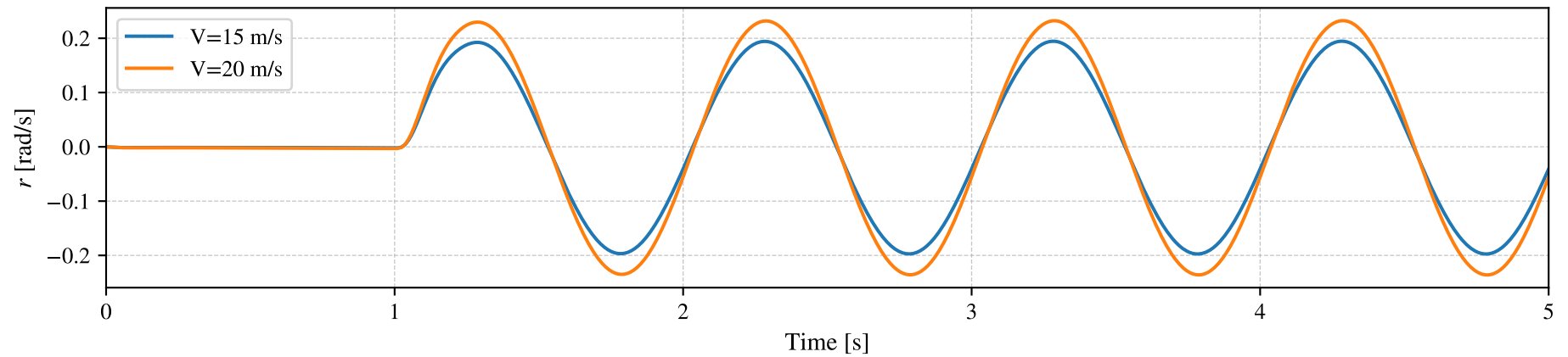
Steering Wheel Angle δ_H



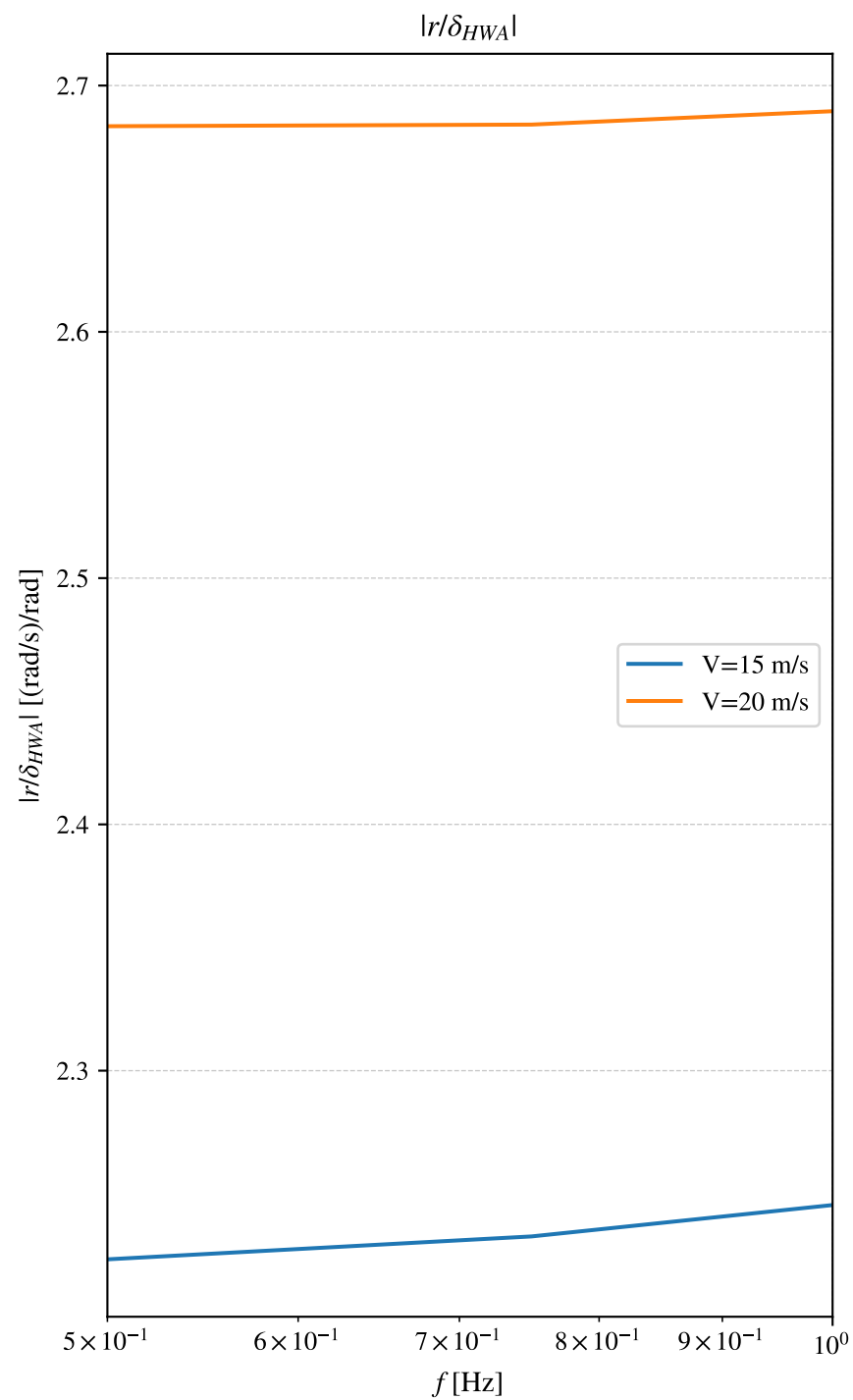
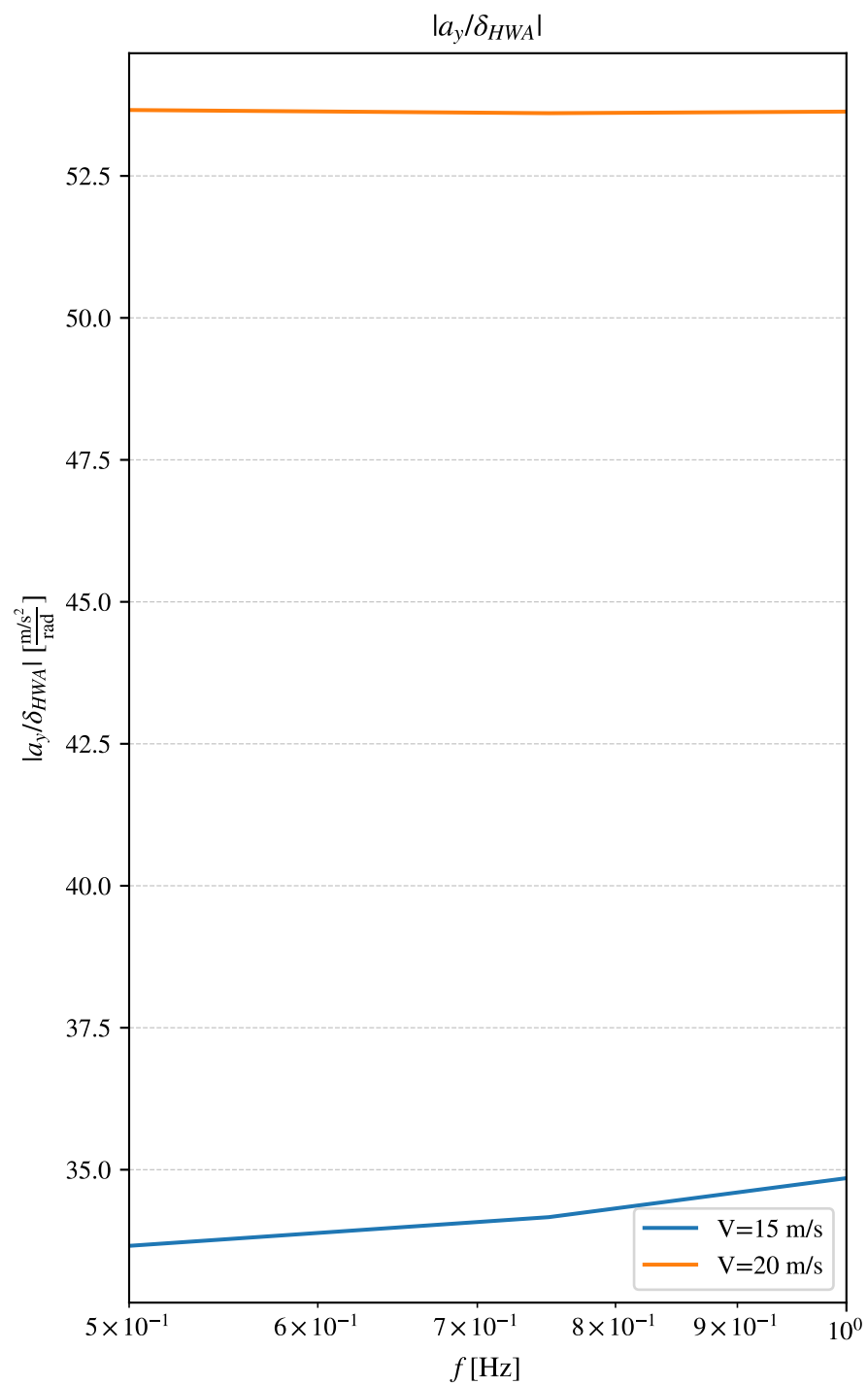
Lateral Acceleration a_y



Yaw Velocity r

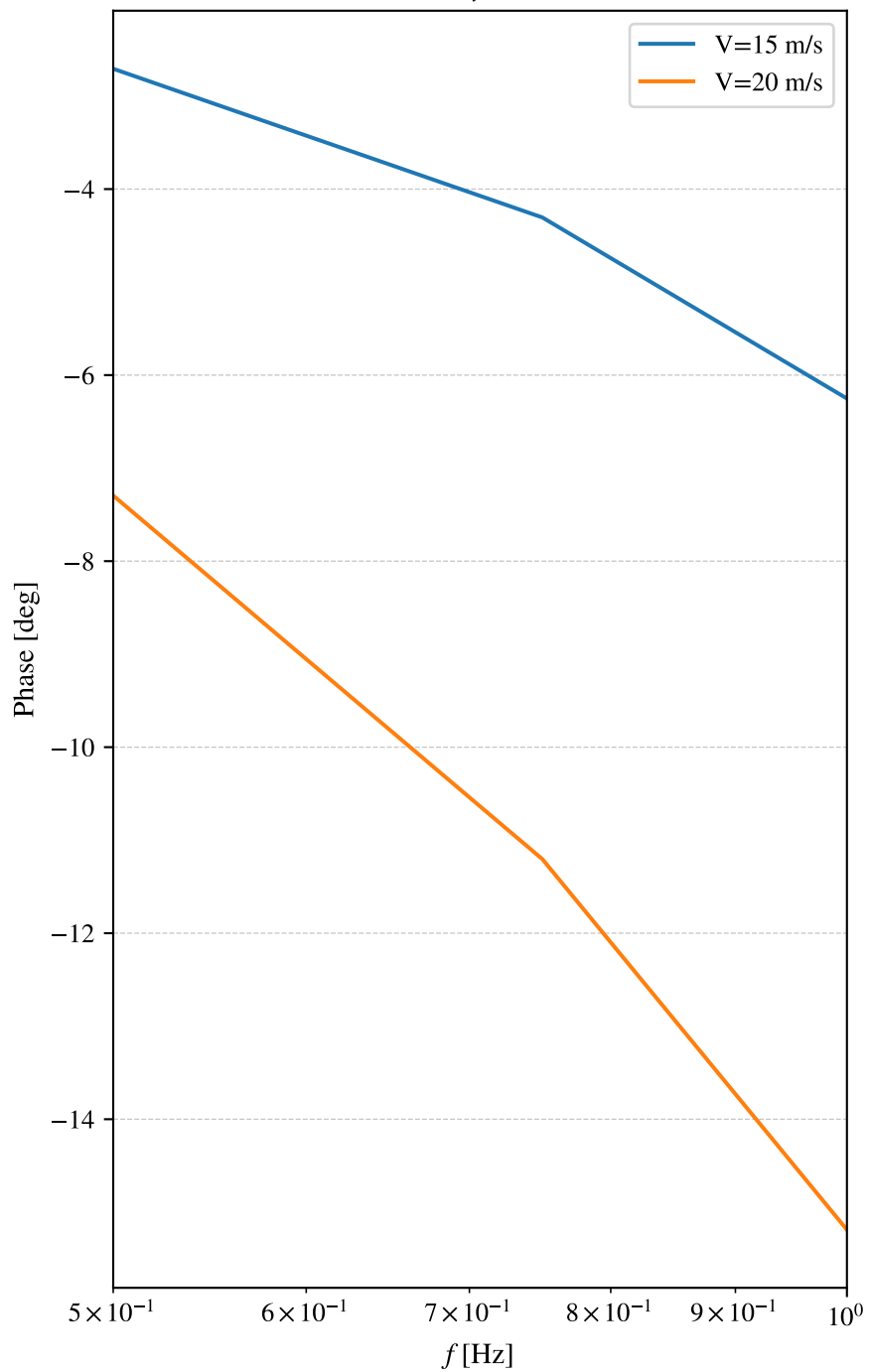


Frequency Response — Gain (TransientEval)



Frequency Response — Phase (TransientEval)

Phase: a_y vs δ_{HWA}



Phase: r vs δ_{HWA}

